Practical No.2

Aim: RSA Encryption and Decryption: Implement the RSA algorithm for public-key encryption and decryption, and explore its properties and security considerations.

```
from Crypto.PublicKey import RSA
from Crypto.Cipher import PKCS1_OAEP
import binascii

# Generate a new RSA key pair
keyPair = RSA.generate(1024)

# Extract the public key
pubKey = keyPair.publickey()
print(f"Public key: (n={hex(pubKey.n)}, e={hex(pubKey.e)})")

# Export the public key to PEM format
pubKeyPEM = pubKey.exportKey()
print(pubKeyPEM.decode('ascii'))

# Extract the private key
privKey = keyPair
print(f"Private key: (n={hex(privKey.n)}, d={hex(privKey.d)})")

# Export the private key to PEM format
privKeyPEM = privKey.exportKey()
print(privKeyPEM.decode('ascii'))

# Encryption
msg = b'SDSM College'  # Message in bytes
encryptor = PKCS1_OAEP.new(pubKey)
encrypted = encryptor.encrypt(msg)
print("Encrypted:", binascii.hexlify(encrypted))

# Decryption
decryptor = PKCS1_OAEP.new(privKey)
decrypted = decryptor.decrypt(encrypted)
print("Decrypted:", decrypted.decode('utf-8'))
```

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Output:

```
============================== RESTART: E:\INS\RSA.py =============
Public key: (n=0xc319cd7997fad7b735ec01490c6a824d8a3d886d6b5e509f
2c4cc15eee025c963a5cfcc6a4990c531df7f3e1768d8e202424561ed91ebc3af
01)
-----BEGIN PUBLIC KEY-----
MIGfMA0GCSqGSIb3DQEBAQUAA4GNADCBiQKBgQDDGc15l/rXtzXsAUkMaoJNij2I
bWteUJ/O+pU+phMXUzh3tvzbYZ2uGfgV5CziP4sXEc8r6xFyEufPI+V95yxMwV7u
AlyWOlz8xqSZDFMd9/Phdo2OICQkVh7ZHrw6/H8QP/i/hB48v2pyN+N3G/7bz01q
PcvQDIMUaC2QtU+PMQIDAQAB
-----END PUBLIC KEY-----
Private key: (n=0xc319cd7997fad7b735ec01490c6a824d8a3d886d6b5e509
72c4cc15eee025c963a5cfcc6a4990c531df7f3e1768d8e202424561ed91ebc3a
4b6e29cf3e81abf97fe9f9ca194946b0ae86fc0a393f1244d015b55678bd5c836
e3a8f1e9c38ca2590236568ae79028f28c9db815c32b5e02a8ed1367401122abb
-----BEGIN RSA PRIVATE KEY-----
MIICWgIBAAKBgQDDGc15l/rXtzXsAUkMaoJNij2IbWteUJ/O+pU+phMXUzh3tvzb
YZ2uGfgV5CziP4sXEc8r6xFyEufPI+V95yxMwV7uAlyWOlz8xqSZDFMd9/Phdo2O
ICQkVh7ZHrw6/H8QP/i/hB48v2pyN+N3G/7bz01qPcvQDIMUaC2QtU+PMQIDAQAB
An83S24pzz6Bq/l/6fnKGUlGsK6G/Ao5PxJE0BW1Vni9XINro3EjLfktQmfUsA+W
uqSUrFsLF3f2wEiA+uRlWMo06Z9AknyWNE46jx6cOMolkCNlaK55Ao8oyduBXDK1
4CqO0TZ0ARIqu7mLtO7IgEj85oFomvc3u0zCslBPaDFBAkEA0Lnvc2sy57Q6V4zO
4Y/6nbX9S+gifrNfza+HcIY2tYX+t9953TFtXsUKl4bTTuMth431yPJQTLowLLh6
KDeZuQJBAO9J2jKPNcLFBrvhF16rGXu8RWvXoHfFol/+8aXZPzbxRCmkqtD740cn
XE2FpJ9kDOP/cVI0rxPQkFlh2sVXfTkCQQDPRiW0r1qw37gkIw96zZjprsUn8RD4
UzarMklEg/OmVbc6b3E8KA5dUz0OdnQ23cDj4SSX9ZGqcCbQ8i1U7Mn5AkBTNBbO
5cUHyIKi6Y26uTvPLpPk/8EoZGwlfPyBcpzsec/9uug6apuIrkMGIaYwVhW1YakB
sJgAB9FpgwETBWzhAkAsjBi4JsJM3e3W1qEcw8QXH7XHQIz+PhmBRngRXITkoeAc
0xJl3dVIvxFS8stp1ikmgTEXtg1+Ss9NNoJsvgP3
-----END RSA PRIVATE KEY-----
Encrypted: b'8b4d057f3b80a7d52d0e059eeaebb2c3b89698562d49ca0b5353
ff350a72bb5de321af87779b2f2a73a681505eb859d6acebe2ca274e26c0526ab
Decrypted: SDSM College
```